MIL-C-915/29B
23 August 1976
SUPERSEDING
MIL-C-915/29A
1 August 1972

MILITARY SPECIFICATION SHEET

# CABLE, ELECTRICAL, 1000 VOLTS, TYPE DSGU

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the cable described herein shall consist of this document and the latest issue of Specification MIL-C-915.

REQUIREMENTS:
Qualification required.

## Construction (Watertight)

### Sizes 3 to 23 inclusive, extruded silicone rubber insulation

First - Copper conductor, uncoated (see Details for size).
Second - Extruded silicone rubber insulation (see Details for diameter.)
Third - Glass braid.
Fourth - Braid covering (see Details for diameter). Standard Identification Code, applied by method 1.
Fifth - The two conductors shall be cabled together with a lay not greater than 24 times the pitch diameter. Either non-fibrous or a combination of non-fibrous and fibrous fillers shall be used to form a firm, well-rounded assembly.
Sixth - Binder or combination binder/barrier.
Seventh - Special thermoplastic jacket (gray) (see Details for thickness). Cable surface marking required.

### Sizes 50 to 400 inclusive, silicone rubber-glass tape insulation

First - Copper conductor, uncoated (see Details for size).
Second - Silicone rubber-glass tape insulation (see Details for diameter). Letter Identification Code, applied by method 5.
Third - The two conductors shall be cabled together with a lay not greater than 24 times the pitch diameter. Either non-fibrous or a combination of non-fibrous and fibrous fillers shall be used to form a firm, well-rounded assembly.
Fourth - A binder.
Fifth - Special thermoplastic jacket (gray). (See Details for thickness). Cable surface marking required.

Note 1: When this cable is manufactured with armor, the following additional requirements shall apply:

(a) The type designation shall change to DSGA.
(B) (b) Standard thermoplastic jacket (black or gray) may be used.
(c) Braided metal armor and paint shall be added.
(d) Cable surface marking will not be required.
(B) (e) The maximum overall dimension shall be increased by 0.050-inch.
(f) Group C inspection shall include: Armor - Conformance to material, construction and coverage. When standard thermoplastic jacket is used, the air oven test does not apply.

(B) denotes changes



Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Ship Engineering Center, Center Building, SEC 6124, Prince George's Center, Hyattsville, Maryland 20782 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6145

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Details

| Type and size | Conductor size Navy Standard | Diameter over outer insulating tape nominal (nom) (inch) | Diameter over insulating minimum (min) (inch) | Diameter over braid covering maximum (max) (inch) | Jacket thickness nom (inch) | Overall diameter max (inches) | Conductor resistance per 1000 ft at 25°Celsius (C), max (ohms) | Insulation resistance per 1000 ft min (megohms) | Cold bending mandrel diameter max (inches) | Accelerated service loading (amps) |
|---|---|---|---|---|---|---|---|---|---|---|
| DSGU-3 | 3(7) | - | 0.096 | 0.130 | 0.030 | 0.391 | 4.0 | 500 | - | - |
| DSGU-4 | 4(7) | - | .112 | .143 | .030 | .427 | 2.51 | 500 | - | - |
| DSGU-9 | 9(7) | - | .154 | .187 | .040 | .544 | 1.25 | 500 | - | - |
| DSGU-14 | 14(7) | - | .230 | .262 | .040 | .670 | 0.785 | 500 | - | 100 |
| DSGU-23 | 23(7) | - | .284 | .316 | .050 | .815 | .496 | 500 | - | 130 |
| DSGU-50 | 50(19) | 0.334 | - | - | .050 | .911 | .23 | 200 | 12 | 215 |
| DSGU-75 | 75(37) | .407 | - | - | .050 | 1.074 | .149 | 175 | 13 | 285 |
| DSGU-100 | 100(61) | .453 | - | - | .050 | 1.167 | .114 | 160 | 15 | 335 |
| DSGU-200 | 200(61) | .634 | - | - | .060 | 1.583 | .057 | 125 | 20 | 525 |
| DSGU-300 | 300(91) | .748 | - | - | .075 | 1.841 | .0382 | 110 | 23 | 700 |
| DSGU-400 | 400(127) | .862 | - | - | .075 | 2.069 | .0273 | 100 | 25 | 835 |



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EXAMINATION AND TESTS:

| | Requirements |
|---|---|
| **Basic electrical:** | |
| Voltage withstand - Volts, root mean square (Vrms), min | |
| Sizes 3 thru 9 | 3000 |
| Sizes 14 and larger | 5000 |
| Insulation resistance - Megohms/1000 ft min | (See Details) |
| Conductor resistance - Ohms/1000 ft max | (See Details) |
| **Group A:** | |
| Visual and dimensional | |
| Watertightness - See MIL-C-915, table of limits for water leakage | |
| Abrasion resistance - Sizes 3 thru 23, scrapes, min | 250 |
| Crack resistance - Sizes 3 thru 23 | No damage |
| **Group B:** | |
| Gas flame - 1 hour | No failure |
| Drip - at 95°C ± 1°C | Zero |
| Cold bending, cable, sizes 50 and larger, - (see Details for mandrel diameter) | No damage |
| Physicals (unaged) | |
| Insulation (extruded) | |
| Tensile strength - pounds-force per square inch ($lbf/in^2$), min | 750 |
| Elongation - Percent, min | 125 |
| Jacket (Standard thermoplastic) | |
| Tensile strength - $lbf/in^2$, min | 1800 |
| Elongation - Percent, min | 225 |
| Jacket (Special thermoplastic) | |
| (B) Tensile strength - $lbf/in^2$, min | 2300 |
| (B) Elongation - Percent, min | 250 |
| **Group C:** | |
| Physicals (aged) | |
| Jacket (Standard thermoplastic) | |
| Hot oil immersion | |
| Tensile strength - $lbf/in^2$, min | 1400 |
| Elongation - Percent, min | 50 |
| Jacket (Special thermoplastic) | |
| (B) Air oven (168 hours at 136°C) | |
| (B) Tensile strength - Percent of unaged, min | 80 |
| (B) Elongation - Percent of unaged, min | 60 |
| (B) Hot oil immersion | |
| (B) Tensile strength - Percent of unaged, min | 75 |
| (B) Elongation - Percent of unaged, min | 50 |
| Permanence of printing (conductor) | |
| Sizes 3 thru 23 - cycles, min | 50 |
| Permanence of printing (jacket) - cycles, min | 250 |
| Cable filler removability | |

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of Groups A, B, and C, plus the following:

| | |
|---|---|
| Cable aging and compatibility (125°C ± 5°C) | No failure |
| Cold working (minus 20°C ± 2°C) | No damage |
| Gas flame - 3 hours | No failure |
| Specific gravity of extruded silicone insulation, max | 1.55 |
| (B) Accelerated service - Sizes 14 and larger (see Details for load current) | No failure |



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UNIT ORDERING LENGTHS:

| Type and size | Feet (nom) |
|---|---|
| DSGU-3 | 2500 |
| DSGU-4 thru 23 | 2000 |
| DSGU-50 | 1500 |
| DSGU-75 and 100 | 1000 |
| DSGU-200 thru 400 | 800 |

Custodians:
Army - MI
Navy - SH
Air Force - 99

Review activities:
Army - EL, AV
Navy - EC

User activities:
Army - ME, MU, WC
Navy - CG

Preparing activity:
Navy - SH

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OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

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MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

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   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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